CHEONGJAE LEE
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EDUCATION

BOSTON UNIVERSITY — Boston, MA
Master's in Electrical Engineering — Expected Graduation: 12/27

THE PENNSYLVANIA STATE UNIVERSITY — State College, PA
*Bachelor of Science in **Electrical Engineering**, Minor in Engineering Mechanics* — 12/23

PROJECTS

MANIMO: GRIP-FORCE-LIMITED GRASPING FOR A MOBILE MANIPULATOR — Boston University
EC601 Team Project, team of 2 (in progress) — Fall 2026 – Present

- Added grip-force limits to robot grasping in Isaac Lab and trained reinforcement learning policies on a GPU cluster.
- Trained policies that lift most objects without crushing them in simulation, and found the original evaluation overstated crushing.
- Built camera perception that estimates an object's type, position, size, and shape in simulation.
- Set up Isaac Sim on BU's GPU cluster, and got the real LanderPi robot driving over ROS 2.
- Wrote an autonomous pick-and-place mission for a simulated LanderPi, using the real robot's ROS 2 interfaces.
- Introduced a rule that every arm command is checked in simulation before it runs on the real robot.

MULTI-CROP DETECTION FOR AN AGRICULTURAL ROBOT — Boston University
Robotics Club, Agrobot Team (in progress) — Fall 2026 – Present

- Trained a tomato detection model (Faster R-CNN) in PyTorch for the club's agricultural robot.
- Built the dataset pipeline and compared model settings to choose the fastest accurate model.
- Flagged that the detector can mistake human hands for tomatoes, so it must not be used as a safety check.

PENN STATE ADVANCED VEHICLE TEAM (AUTODRIVE CHALLENGE II) — State College, PA
Head of Vehicle Systems (ME496) and Electrical Engineer (ME440W Capstone) — 01/23 – 12/23

- Designed the electrical power distribution, sensor suite, and roof rack for an autonomous Chevrolet Bolt EUV.
- Led vehicle-systems hardware: waterproofed sensors and harnesses, and integrated sensors with the central compute.

PROFESSIONAL EXPERIENCE

SOELECT INCORPORATED — Greensboro, NC
Manufacturing Process Engineer — 03/26 – 05/26

- Ran roll-to-roll lithium metal anode production and tested battery cells for performance and quality.
- Assembled pouch cells and inspected anode quality with microscopy and adhesion testing.

FASTLION ENERGY INCORPORATED — State College, PA
Electrical Engineer — 07/24 – 03/26

- Designed and prototyped battery management systems (BMS) for military and industrial batteries.
- Designed multi-layer PCBs for precision sensing, power control, and high-current paths.
- Developed CAN communication so the battery reports its status and faults to the vehicle.
- Tested, debugged, and validated prototype hardware with standard lab equipment.
- Worked with mechanical and systems engineers to meet performance, safety, and environmental requirements.

EC POWER GROUP — State College, PA
Electrical Engineer (promoted from Electrical Engineering Intern, 02/24) — 05/23 – 07/24

- Led end-to-end development of a battery test platform: circuit boards, C/C++ firmware, and data collection.
- Programmed BMS prototypes and built an automated battery test rig.

SKILLS

Programming: Python, C/C++, MATLAB, LabVIEW
Robotics and ML: ROS 2, Isaac Sim, Isaac Lab, reinforcement learning (PPO), PyTorch, RGB-D perception
Tools and Hardware: Linux, Git, HPC (Singularity), Altium, KiCad, CAN, SPI, I2C, oscilloscopes, logic analyzers
Lab and Mechanical: SEM/EDX, EIS, li-ion battery cycle testing, SolidWorks, OnShape, 3D printing